INCH-POUND
MIL-B-8952B
15 JANUARY 1995
SUPERSEDING
MIL-B-8952A
19 November 1976

MILITARY SPECIFICATION

BEARING, ROLLER, ROD END, ANTIFRICTION SELF-ALIGNING

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers antifriction self-aligning rod end roller bearings operating in the temperature range of -67° to +300° F (-55° to +149° C) and intermittent spikes to 350° F (177° C).

1.2 Classification. Bearings shall be of the following types, as specified (see 6.2.1).

Type I - External thread rod end bearings

Type II - Internal thread rod end bearings

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, System Standardization, Code 4.1.11.2 B120-3, Naval Air Warfare Center Aircraft Division, (Highway 547) Lakehurst, NJ 08733-5100 by using the self-addressed Standardization Document Improvement Proposal (dd Form 1426) appearing at the end of this document or by letter.

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SPECIFICATIONS

FEDERAL

QQ-P-416 – Plating, Cadmium (Electro Deposited)

MILITARY

MIL-P-197 – Bearings, Antifriction, Associated Parts and Subassemblies Packaging of
MIL-S-8879 – Screw Threads, Controlled Radius Root With Increased Minor Diameters; General Specification For
MIL-G-23827 – Grease, Aircraft and instrument, Gear and Actuator Screw
MIL-G-81322 – Grease, Aircraft, General Purpose Wide Temperature Range

STANDARDS

MILITARY

MIL-STD-105 – Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 – Marking for Shipment and Storage
MIL-STD-130 – Identification Marking of US Military Property
MIL-STD-1949– Inspection, Magnetic Particle

(Unless otherwise indicated, copies of the federal and military specifications, standards, and handbooks are available from the DODSSP-Customer Service, Standardization Documents Order Desk, 700 Robbins Ave, Bldg 4D, Philadelphia, PA 19111-5094.

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

CONSOLIDATED CLASSIFICATION COMMITTEE

Uniform Freight Classification Rules

(Application for copies of the above publication should be addressed to the Consolidated Classification Committee, 202 Chicago Union Station, Chicago, IL 60606.)

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2.2 Non-Government publications. The following document forms a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of the documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI/ASME B46.1 Surface Texture (Surface Roughness, Waviness and Lay)

(Applications for copies of the above publication should be addressed to the American National Standard Institute, 11 West 42nd Street, New York, NY 10036.)

AEROSPACE INDUSTRY ASSOCIATION OF AMERICA (AIA)

NAS 516 - Fitting, Lubrication - 1/8 Inch Drive, Flush Type

(Application for copies of the above publication should be addressed to the AEROSPACE INDUSTRY ASSOCIATION OF AMERICA 1250 EYE ST. N.W. Washington, DC 20005.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

* ASTM A380 Practice for Cleaning and Descaling Stainless Steel Parts, Equipment, and Systems

(Application for copies should be address to the American Society for Testing and Materials, 1916 Race St., Philadelphia, PA 19103-1187)

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

* AMS 2417 Plating, Zinc-Nickel Alloy

(Application for copies should be addressed to SAE International, 400 Commonwealth Drive, Warrendale, PA 15096-0001.)

(Non-Government standards and other publications are normally available from organizations that prepare the documents. These documents also may be available in or through libraries or other informational services.)

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2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, (except for the related associated Military Specifications Sheet) the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Military Specification sheets. The individual item requirements shall be as specified herein and in accordance with the applicable Military Specifiction sheets. In the event of any conflict between requirements of this specification and the Military Specification sheets, the latter shall govern.

3.1.1 Product change. Any change in product design, including raceway geometry or dimensions, rolling element dimensions, rolling element quantity, seals, materials or plant location shall be reported to the qualifying activity and will require requalification of the product to an extent determined by the qualifying activity. Any other other specific changes which must be brought to the attention of the qualifying activity will be identified in the qualification notification letter.

3.1.2 Product manufacture. The bearing manufacturer shall be capable of performing the preponderance of manufacturing operations in-house, but may subcontract these operations at its option. Component inspection shall be performed at the plant listed on the Qualified Products List (QPL). If manufacturing operations are performed in more than one plant, the manufacturer's additional plant(s) shall be reported to the qualifying activity and will be listed accordingly on the QPL. The manufacturer is responsible for meeting all requirements of the specification and for the quality of the end product, whether it is manufactured totally in-house or some of the operations are performed by a subcontractor. Inherent in the responsibility for the end product is the responsibility to verify that the subcontractor's processes meet specification requirements. (NOTE: Bearings partly or completely manufactured in foreign countries shall be subject to the laws and procurement regulations pertaining to acquisition of foreign made products.)

3.2 Qualification. The bearings furnished under this specification shall be a product which has been subjected to and which has passed the qualification tests specified herein, and which has been listed on or approved for listing on the applicable Qualified Products List.

3.3 Material. The bearings shall conform to the applicable Military Specification sheet.

3.4 Design and construction. Only factory new bearings conforming to requirements specified herein shall be furnished.

3.4.1 Hardness. The inner ring and rollers shall be through-hardened and shall have a hardness of Rockwell C60 to C65. The outer raceway be through-hardened or case-hardened and shall have a minimum hardness of Rockwell C60. Total case depth shall not exceed 50 percent of cross sectional area. Rod end shanks shall have a uniform cross sectional hardness with minimum hardness of Rockwell C27.

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3.4.2 Surface roughness. The sides of the inner ring shall have a smooth, machined finish that will not exceed 125 microinches Ra. Otherwise, the surface roughness shall be as specified on the Military Specification Sheets.

3.4.3 Seals. Seals of contact type shall be provided to prevent entrance of dust and to retain the lubricant throughout the temperature range specified. The material shall be reinforced TFE fiberglass.

3.4.4 Relubrication provisions. Provisions for lubrication shall be made by use of removable seals or by use of an external lubrication fitting as specified on the Military Specification Sheets. When removable seals are used, they shall be readily removable without damage, and shall be of such construction that they may be reused without undue difficulty. When external lubrication fittings are provided, they shall conform to NAS516-1 as shown on the applicable drawing.

3.4.5 Retainers. Roller retainers shall be made of steel or beryllium-copper in accordance with applicable Military Specification Sheets.

3.4.6 Rings. The outer rings shall be of single piece construction with the rod end shank. The inner ring shall be of single piece construction.

3.4.7 Threads. Threads shall conform to UNJF-3A and UNJF-3B in accordance with MIL-S-8879 for external and internal threads, respectively.

* 3.4.8 Lubrication. The bearings shall be thoroughly cleaned and dried before being lubricated. The bearings shall be filled with grease conforming to MIL-G-81322 or MIL-G-23827 so that at least eighty percent of the bearing void shall be filled. Bearings shall not have been lubricated more than 24 months prior to date of delivery. The bearing shall be preserved with the operating lubricant.

3.4.9 Dimensions and tolerances. Dimensions and tolerances shall be in accordance with the applicable Military Specification Sheets.

3.4.9.1 Shank alignment. The point representing the center of self-alignment motion shall be coincident with the axis of the rod end within .010 inch.

3.4.10 Stabilization. When tested as specified in 4.5.15, bearings that show a dimensional change from the previous reading of more than .0001 inch per inch diameter shall be considered to have failed.

3.5 Performance.

3.5.1 Radial strength.

3.5.1.1 Radial limit load rating. When tested as specified in 4.5.6 the bearing shall have a radial limit load rating of not less than the value specified on the Military Specification sheet. If starting torque increases by 100% or more, bearing shall be considered to have failed.

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3.5.1.2 Radial fracture load. When tested as specified in 4.5.7, The minimum static fracture capacity shall be not less than 1.5 times the radial limit load rating specified on the applicable Military Specification sheet. Following the test, bearings that cannot be turned by hand, or that have fractured parts, shall be considered to have failed.

3.5.1.3 Radial dynamic load rating. The bearings shall have a radial dynamic load rating of not less than the values specified on the applicable Military Specification sheet. The average life shall be a minimum of 10,000 cycles oscillating through an arc of 90 degrees and shall be tested as specified in 4.5.8. Failure is defined as a minimum of .01 square inches of spalled race surface. Data is to be Weibull plotted for evaluation. If the B-50 life of the bearings is less than 10,000 cycles, the bearings shall be considered to have failed the test.

3.5.2 Axial strength.

3.5.2.1 Axial limit load rating. When tested as specified in 4.5.9, the bearing shall have an axial limit load rating of not less than the value specified on the applicable Military Specification sheet. If starting torque increases by 100% or more, the bearings shall be considered to have failed.

3.5.2.2 Axial fracture load. After being tested as specified in 4.5.10, bearings that cannot be turned by hand or that have parts fractured, shall be considered to have failed.

3.5.3 Radial internal clearance. When tested as specified in 4.5.11, radial internal clearance measured before the radial limit load rating test (4.5.6) shall be as specified on the applicable Military Specification sheet.

3.5.4 Axial internal clearance. When tested as specified in 4.5.12, axial internal clearance measured before the axial limit load rating test (4.5.9) shall be as specified on the applicable Military Specification sheet.

3.5.5 Seal performance.

3.5.5.1 Lubricant retention. Bearings shall withstand the lubricant retention test specified in 4.5.5. without failure. No less than 50 percent of the lubricant's weight shall be retained in the bearing at the end of the test.

* 3.5.5.2 Dust. Bearings subjected to the dust test procedure specified in 4.5.14 shall meet the following requirements:

a. Starting torque increase no more than 100 percent over initial measured value.

b. Radial internal clearance increase no more than 50 percent over initial measured value.

c. There shall be no deterioration of the seals or ingress of dust into the bearing.

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3.6 Weight. The weight of the bearing shall be as specified on the applicable Military Specification sheet.

3.7 Plating. Plating, when specified, shall be zinc-nickel alloy plated in accordance with AMS 2417, Type 2, or cadmium plating in accordance with QQ-P-416, Type I, class 2, to a thickness of .0003 to .0006 inch. Plating shall be on all external steel surfaces, except bore, seals and seal retainers.

3.8 Identification of product. Bearings shall be permanently and legibly marked for identification in accordance with MIL-STD-130 on outer race.

3.9 Workmanship. All details of workmanship shall be high grade to ensure satisfactory service life. Parts shall not contain sharp edges, burrs, loose chips, dirt, or other foreign matter. The mating surfaces of the rod end the inner ring with rollers shall be free of tool marks, galling, and scratches after assembly.

3.10 Magnetic particle inspection. Certification of such inspection, in accordance with 4.5.1.2, shall be furnished with Qualification and Quality conformance inspection samples.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of Sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with requirements of the contract. Sampling inspection, as a part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The examination and testing of the bearings shall be classified as:

a. Qualification (4.3).

b. Quality conformance inspection (4.4).

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TABLE I. QUALIFICATION INSPECTION SAMPLES.

	SAMPLE NUMBER																										
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	REQUIREMENT	TEST METHOD
EXAMINATION	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	3.1, 3.3, 3.4, 3.6 3.7, 3.8, 3.9	4.5.1
PACKAGING	X	X	X																								4.5.1.1
HARDNESS	X	X	X																							3.4.1	4.5.2
SURFACE ROUGHNESS	X	X	X																							3.4.2	4.5.3
PLATING	X	X	X																							3.7	4.5.4
LUBRICATION	X	X	X																							3.4.8	4.5.5
RADIAL INTERNAL CLEARANCE				X	X	X	X	X	X	X	X	X				X	X	X	X	X	X	X	X	X	X	3.5.3	4.5.11
AXIAL INTERNAL CLEARANCE				X	X	X							X	X	X	X	X	X	X	X	X	X	X	X	X	3.5.4	4.5.12
STABILIZATION				X	X	X																				3.4.10	4.5,15
LUBRICANT RETENTION				X	X	X																				3.5.5.1	4.5.13
DUST							X	X	X																	3.5.5.2	4.5.14
RADIAL LIMIT LOAD										X	X	X														3.5.1.1	4.5.6
RADIAL FRACTURE LOAD										X	X	X														3.5.1.2	4.5.7
AXIAL LIMIT LOAD													X	X	X											3.5.2.1	4.5.9
AXIAL FRACTURE LOAD													X	X	X											3.5.5.2	4.5.10
RADIAL DYNAMIC LOAD																X	X	X	X	X	X	X	X	X	X	3.5.1.3	4.5.8
MAGNETIC PARTICLE	X	X	X																							3.10	4.5.1.2

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4.3 Qualification inspection. Qualification inspection shall consist of all test specified under 4.5. The test shall be conducted as specified in table I.

4.3.1 Qualification Test Samples. Test samples shall consist of 25 bearings for each Military Specification Sheet dash number, representative of the production bearing. The samples shall be identified with the manufacturer's part number and such other information as required by the procuring activity (see 6.2).

4.3.2 Test reports. The manufacturer's test report shall be prepared in accordance with the qualifying activity authorization letter. The report shall include a dated drawing which shall describe in detail the manufacturer's product by specifying all dimensions and tolerances, materials, construction, lubricant, plating, and heat treatment. Any information the manufacturer deems proprietary must be so indicated.

4.3.3 Qualification failure. For intial qualification, any failures during examinations or tests shall be reported to the qualifying activity before continuing testing. Based on corrective action, authorization to complete the qualification test may be granted by the qualifying activity. The applicant's products shall pass all requirements of the specification. Failure of any requirement of this specification shall be cause for refusal to grant quailfication.

4.3.4 Retention of qualification. In order to retain qualification, the results of the retention of qualification tests, listed in table II, shall be furnished to the qualifying activity at 36 month intervals. One part number from each military specification sheet for which the manufacturer is qualified shall be tested. The qualifying activity shall select the dash number to be tested. To retain qualification, the manufacturer's products shall pass all of the retention of qualfication tests. Failure to furnish the test information shall result in loss of qualification for that product.

4.3.4.1 Failure of retention of qualification. Any failures during retention of qualification testing shall be reported immediately to the qualifiying activity. Authorization to continue retention of qualification testing will be based on acceptance of the corrective action.

Table II. Retention of Qualification examinations and test samples.

Examination or test	Sample number 1 2 3 4 5 6	Requirement paragraph	Test method paragraph
Examination of product	X X X X X X	3.1, 3.3, 3.4 3.6, 3.7, 3.8 3.9	4.5.1
Hardness	X X X	3.4.1	4.5.2
Surface roughness	X X X	3.4.2	4.5.3
Plating	X X X	3.7	4.5.4
Lubrication	X X X	3.4.8	4.5.5
Radial internal clearance	X X X	3.5.3	4.5.11
Axial internal clearance	X X X	3.5.4	4.5.12
Stabilization	X X X	3.4.10	4.5.15

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4.4 Quality conformance inspection. Quality conformance inspection shall consist of:

a. Individual tests.

b. Sampling tests.

4.4.1 Individual tests. The individual tests shall consist of the examination of product.

4.4.2 Sampling tests. The sampling tests shall consist of the tests of 4.5, except 4.5.6, 4.5.7, 4.5.8, 4.5.9, 4.5.10, 4.5.13, 4.5.1.4, and 4.5.15

4.4.2.1 Lot. The lot definition, formation, and size shall be in accordance with MIL-STD-105.

4.4.2.1.1 Samples. Sample bearings shall be selected in accordance with inspection level II of MIL-STD-105, the lot shall be rejected if one or more defects are found.

4.5 Test methods.

4.5.1 Examination of product. The bearings shall be carefully examined for conformance to this specification and the applicable Military Specification sheets with respect to dimensions, identification, workmanship, marking, shank alignment, lubrication provisions and threads.

4.5.1.1 Preservation, packaging, and packing. Preservation, packaging, and packing shall be examined for conformance with section 5.

4.5.1.2 Magnetic particle inspection. This inspection shall be in accordance with MIL-STD-1949. The bearing outer race and shank shall be 100% magnetically inspected by both longitudinal and circular methods prior to assembly. (See 3.10)

4.5.2 Hardness. The bearings shall be disassembled and tested for hardness, using a standard Rockwell hardness tester. The test shall be made on the rolling contact area of the inner and outer raceways and on the rollers. Readings for rollers shall be taken on a flat surface. Readings for rod ends shall be taken on or near the end of the shank. (See 3.4.1)

4.5.3 Surface roughness. Surface roughness shall be measured in accordance with ANSI/ASME B46.1. The test shall be made on the bearing disassembled in 4.5.2. (See 3.4.2)

4.5.4 Plating. Plating shall be tested for comformance with 3.7 and the applicable Military Specification sheet.

4.5.5 Lubrication. The lubricant shall be inspected for quality and cleanliness, (See 3.4.8).

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4.5.6 Radial limit load. The bearings shall be mounted in a rigid support as shown on figure 1, and the bearing rotated manually to determine operational smoothness before testing. The starting friction torque shall be recorded. The appropriate load specified on the applicable Military Specification sheet shall be applied for one minute. The load shall be removed and the bearing inspected for evidence of failure. (See 3.5.1.1)

4.5.7 Radial fracture load. The radial fracture load of 1.5 times the radial limit load specified on the Military Specification sheet shall be applied to the bearing for one minute. After removal of the load, the bearing shall be examined for failure. Bearings of each Military Specification part number submitted shall be tested. (See 3.5.1.2)

4.5.8 Radial dynamic load. Ten bearings of each Military Specification Sheet part number submitted shall be tested for radial dynamic load. The bearings shall be mounted on a test fixture having suitable support bearings and drive linkages to provide oscillation of the inner ring. The radial dynamic load specified on the Military Specification Sheet shall be applied unidirectionally with respect to the outer ring and maintained constant within ±1 percent. Each bearing shall be oscillated through an arc 90 degrees and back to the starting position. Termination of each test shall be an indication of spalled race surfaces as measured by roughness of operation or increase increase in radial internal clearance, either method established to yield about 90 percent failed samples. The bearings shall be disassembled and the raceways inspected. (See 3.5.1.3)

4.5.9 Axial limit load. The bearing shall be mounted in rigid support as shown on figure 1 and the bearing rotated manually to determine operational smoothness before testing. The starting friction torque shall be measured. The appropriate load specified on the applicable Military Specification sheet shall be applied for one minute. The load shall be removed and the bearing inspected for evidence of failure. (See 3.5.2.1)

4.5.10 Axial fracture load. The axial fracture load of 1.5 times the axial limit load specified on the Military Specification sheet shall be applied to the bearing for one minute. After removal of the load, the bearings shall be examined for failure. (See 3.5.2.2)

4.5.11 Radial internal clearance. The bearings of each Military Specification sheet size shall be tested for radial internal clearance. The inner ring shall be held rigidly and a radial load of 22 pounds applied alternately in opposite directions to the outer ring. A dial indicator shall be used to measure total movement of the outer ring. (See 3.5.3)

4.5.12 Axial internal clearance. Bearings of each Military Specification sheet size shall be tested for axial internal clearance. The outer ring shall be held rigidly and an axial load of 22 pounds applied alternately in opposite direction to the inner ring. A dial indicator shall be used to measure total movement of the inner ring. (See 3.5.4)

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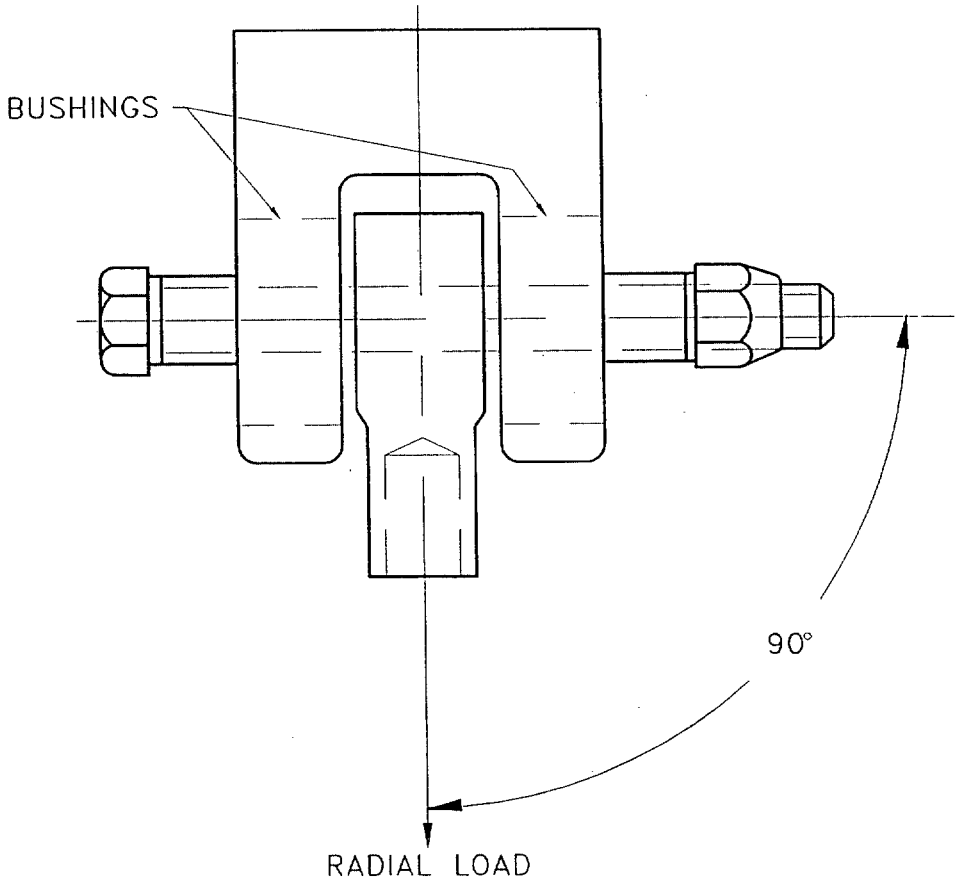


Figure. 1. Radial Strength Test Setup

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4.5.13. Lubrication retention test. Three bearings of each size shall be maintained at a maximum of -65°F (-54°C) for 100 hours and then at a minimum of 300°F (149°C) for 100 hours while being operated in the test chamber at 3 revolutions per minute (RPM) with no applied bearing load. The bearings shall be mounted on a horizontal shaft with the outer rings held stationary. (See 3.5.5.1)

4.5.14 Dust test Three bearings of each military specification sheet size shall be subjected to the dust test. Bearings shall be tested for 48 hours 77°F ±5°F (25°C ±3°C) in the closed test chamber specified in 4.5.14.1. The bearings shall be mounted on a horizontal shaft and shall be operated at 3 revolutions per minute (RPM) with no applied bearing load and with the outer rings held stationary. Pre-test values of starting torque and radial internal clearance shall be recorded. Post-test measurements of starting torque shall be made in the "as tested" condition. Before post-test measurement of radial internal clearance, seal retainers shall be removed and the bearings cleaned thoroughly and relubricated. Bearings shall be reassembled before radial internal clearance is measured. (See 3.5.5.2)

4.5.14.1. Test chamber requirements. The test chamber shall be cubic, 2 feet per side, and shall provide view ports for test observation. A minimum layer of 1 inch of dust test media shall be maintained on the chamber floor. Test chamber actuation shall provide continuous movement of dust test media, creating a uniform cloud which shall render test specimens not readily visible through the view ports. Test specimens shall have a minimum spacing of 3 inches from each other and from chamber walls, and the mounting shaft center line shall be 8 to 16 inches above the chamber bottom.

4.5.14.2 Dust test media. Dust test media shall be Arizona Road Dust supplied to SAE Coarse Grade.

4.5.15 Stabilization (qualification). The bearing shall be held at 300°F (149°C) ±10°F for 100 hours; allowed to return to room temperature, then measured. The bearings shall then be cold soaked at -67°F (-55°C) +10°F for 100 hours, allowed to return to room temperature, then measured. All measurements shall be made at room temperture and shall be made at a single reference position. (See 3.4.10)

5. PACKAGING

5.1 Preservation. Preservation shall be in accordance with MIL-P-197, Level A, C, or commercial as specified.

5.2 Packing. Packing shall be in accordance with MIL-P-197, Level A, B, C or commercial as specified.

5.3 Marking. Marking shall be accordance with MIL-P-197, and MIL-P-129.

5.4 Navy fire-retardant requirements. Requirements shall be as specified in MIL-P-197.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Sealed and shielded antifriction bearings are intented for use in aircraft control systems. Where the exclusion of dust, salt spray, and other extraneous materials is desirable, contact seals are the preferred type. These bearings are not intented for application involving high rotative speeds.

6.2 Acquisition requirements. Procurement documents should specify:

6.2.1 Procurement requirements.

a. Title, number, and date of this specification.

b. Type and Military Specification Sheet Part No. (see 1.2 and 3.3).

c. Quantity.

d. Data requirements (see 3.2 and 4.3.2).

e. Levels of preservation, packaging, and packing (see 5.1 and 5.2).

6.2.2 Contract data requirements. Data specified in paragraph 4.3.2 will be listed directly on a DD Form 1423 incorporated into the contract.

6.3 Qualification. With respect to products requiring qualification, awards will be made only for such products as have, prior to the time set for opening of bids, been tested and approved for inclusion in the applicable Qualified Products List, whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they proposed to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is the Commander, Naval Air Warfare Center, Aircraft Division, MS-60, 6000 East 21st Street, Indianapolis, IN 46219-2189, and information pertaining to qualification of products may be obtained from that activity.

6.3.1 Qualification of a bearing distributor. Department of Defense standardization regulations have specific requirements covering qualifications approval of a distributor. Details of these requirements may be obtained from the qualifying activity.

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6.3.2 Procurement of bearings from a distributor not listed on the QPL. Government contractors and subcontractors who plan to procure a qualified bearing from a distributor not listed on the QPL should be aware that distributors frequently stock bearings which were manufactured several years earlier. Since specification requirements are occasionally changed without a corresponding change in part marking, the parts offered for sale by the distributor may not meet current specification requirements. To assure that parts meet current specification requirements, the contractor should require the following information from the distributor:

(1) Certification that the bearings have been inspected for compliance to the current specification requirements.

(2) Certification of the orignal manufacturer's name, part number and date of of lubrication on the outside of each unit package.

(NOTE: DOD regulations state that only distributors who been approved by the qualifying activity for listing on a QPL may rebrand a QPL listed item.) item.)

6.4 Dust test media. Dust test media may be purchased from Powder Technologies Inc., P.O. Box 1464, MN 55337, or from other sources.

6.5 Subject term (keyword) listing.
Antifriction
Bearing
Lubricated
Teflon seal
Zinc nickel plating

* 6.6 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationships to the last previous issue.

Custodians:
Navy - AS
Air Force - 99
Army - AV

Review activity:
Air Force - 84
DLA- IS
ARMY - AR, MI, AT

Preparing activity:
Navy - AS

(Project 3110-0900)

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3. DOCUMENT TITLE
BEARINGS, ROLLER, ROD END, ANTIFRICTION, SELF-ALIGNING

4. NATURE OF CHANGE (*Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.*)

5. REASON FOR RECOMMENDATION

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a. NAME (*Last, First, Middle Initial*)	b. ORGANIZATION	
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a. NAME COMMANDER NAVAL AIR WARFARE CENTER AIRCRAFT DIVISION	b. TELEPHONE NUMBER (*Include Area Code*) (1) Commercial (908) 323-2970 (2) DSN 624-2970
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